Challenge: Autonomous Beacon Return

Points: 400 • Estimated Time: 90 Minutes

Challenge Scope

The robot must travel away from a home base, circle a distant pillar, and return autonomously to the starting point.

Instructions & Engineering Hints

• **Mission Objective**: Your challenge is to program your robot to autonomously depart from a designated home base, navigate outward to locate and circle a distant pillar, and then reliably return to its exact starting position without any manual intervention.

• **Architectural Hint - Navigation & Odometry**: Consider how your robot will track its journey. Relying solely on timed movements can lead to drift. Think about how integrating wheel encoders or maintaining a consistent heading using a gyro/compass sensor might help you measure distance and turns more accurately.

• **Architectural Hint - Sensor Placement**: To detect or wrap around the pillar, think carefully about where your sensors are positioned on the chassis. Placing a distance or color sensor off-center can simplify wall-following or orbiting behaviors, but make sure it isn't obstructed by other structural components.

• **Strategic Approach**: Break the challenge down into discrete state machine phases: 1) Exit home base, 2) Transit to the pillar, 3) Orbit or loop the pillar completely, and 4) Return transit and final stop at home base. Test each phase independently before stringing them together into a single autonomous run.